CLASS 9 · CBSE · GANITA MANJARI PART 1 · FORMULA BANK

Chapter 1: Orienting Yourself: The Use of Coordinates

Every formula and result of the chapter on one sheet

1	Origin	$(0, 0)$
2	Point on the x-axis	$(x, 0)$
3	Point on the y-axis	$(0, y)$
4	Signs in the quadrants	I $(+, +)$, II $(-, +)$, III $(-, -)$, IV $(+, -)$
5	Distance between (x_1, y) and (x_2, y)	$\lvert x_2 - x_1 \rvert$
6	Distance between (x, y_1) and (x, y_2)	$\lvert y_2 - y_1 \rvert$
7	Distance formula	$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
8	Distance from the origin	$\sqrt{x^2 + y^2}$
9	Reflection in the x-axis	$(x, y) \to (x, -y)$
10	Reflection in the y-axis	$(x, y) \to (-x, y)$
11	Mid-point	$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$
12	Collinear points	$AB + BC = AC$ for some order of the three points
13	Point on a circle (centre C, radius r)	Its distance from C equals r